

# Sooyong Shin

📍 Daejeon, South Korea    ✉ exceldra5@gmail.com    in exceldra5    🌐 exceldra5

## Research Interests

I am interested in **Computer Vision and Robot Perception**, particularly in building **Spatial Intelligence** that enables robots to understand and interact with complex real-world environments. My research interests span **SLAM**, geometric and semantic scene understanding, and temporally persistent perception. I am particularly interested in integrating **Multimodal and Vision-Language Models** with spatial representations and memory to connect visual perception with context-aware reasoning and reliable robotic action.

**Keywords:** Robot Perception, Spatial Intelligence, SLAM, Vision-Language Models

## Education

|                                                                                     |                             |
|-------------------------------------------------------------------------------------|-----------------------------|
| <b>Korea Advanced Institute of Science and Technology</b>                           | <i>Daejeon, South Korea</i> |
| <i>B.S. Candidate in Electrical Engineering and Computer Science (Double Major)</i> | <i>Feb 2018 – Feb 2027</i>  |
| <i>English: TOEFL iBT 5.5/6.0 (MyBest Scores)</i>                                   |                             |

## Research Experience

|                                                                                                                    |                             |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>NMAIL</b> (Supervisor: Sungho Jo)                                                                               | <i>Daejeon, South Korea</i> |
| <i>Research Intern</i>                                                                                             | <i>Mar 2026 – Present</i>   |
| ◦ Conducting research on <b>robot perception</b> and <b>vision-language-action models</b>                          |                             |
| <b>URL</b> (Supervisor: Hyun Myung)                                                                                | <i>Daejeon, South Korea</i> |
| <i>Research Intern</i>                                                                                             | <i>Jun 2025 – Feb 2026</i>  |
| ◦ Conducted research on <b>robot perception</b>                                                                    |                             |
| <i>Individual Study</i>                                                                                            | <i>Mar 2025 – Jun 2025</i>  |
| ◦ Conducted an individual study on <b>ROS</b> and <b>visual SLAM</b>                                               |                             |
| <b>KC-ML2</b>                                                                                                      | <i>Seoul, South Korea</i>   |
| <i>Research Intern</i>                                                                                             | <i>Aug 2024 – Feb 2025</i>  |
| ◦ <b>Co-first author</b> of a paper currently under review at <b>NeurIPS 2026 Evaluations &amp; Datasets Track</b> |                             |
| ◦ Conducted research on <b>relocalization for robot mapping</b>                                                    |                             |
| ◦ Investigated <b>text-based failure recovery</b> in monocular <b>visual SLAM</b>                                  |                             |
| ◦ Skills: C++, Python, ORB-SLAM3, YOLO, PaddleOCR                                                                  |                             |
| <b>SIIT Lab</b> (Supervisor: Junmo Kim)                                                                            | <i>Daejeon, South Korea</i> |
| <i>Individual Study</i>                                                                                            | <i>Jul 2024 – Aug 2024</i>  |
| ◦ Conducted an individual study on <b>vision-language models</b>                                                   |                             |

## Work Experience

|                                                                                                                                                                                                                                       |                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>Bookend</b>                                                                                                                                                                                                                        | <i>Daejeon, South Korea</i> |
| <i>Software Engineer</i>                                                                                                                                                                                                              | <i>May 2023 – Aug 2024</i>  |
| ◦ Built the frontend and backend of <b>SENTENCIFY</b> <a href="#">🔗</a> ( <a href="#">Chrome Web Store</a> <a href="#">🔗</a> ), a multilingual AI writing and paraphrasing application integrating multiple large language model APIs |                             |
| ◦ Developed the service using <b>React</b> , <b>FastAPI</b> , and <b>MongoDB</b>                                                                                                                                                      |                             |
| <i>Software Engineering Intern</i>                                                                                                                                                                                                    | <i>Dec 2022 – Feb 2023</i>  |
| ◦ Developed <b>TYPHERE</b> , a React Native mobile platform connecting writers and publishers throughout the publishing workflow                                                                                                      |                             |

## Projects

---

### CareON — Safe Medical AI for Post-Discharge AML Care

Jul 2026

*Excellence Award (2nd Place), Medical AI Competition 2026*  

- Served as **AI & Systems**, designing a fail-closed medical AI system for post-discharge AML care.
- Integrated deterministic safety rules, grounded RAG, abstention, and audit logging to support reliable and traceable patient guidance.
- Built the **FastAPI** backend and **Flutter** client, and evaluated the system using synthetic clinical scenarios.

### KAIST Swimming Team KAORI Web Platform [www.kaist-kaori.org](http://www.kaist-kaori.org)

Since 2024

- Developed and maintained a full-stack web platform for managing member profiles and swimming records
- Enabled individual and team performance analysis through structured historical record data
- Skills: React, FastAPI, MongoDB

## Skills

---

**Programming:** Python, C++, JavaScript, TypeScript

**AI & Robotics:** PyTorch, ROS, ORB-SLAM3, YOLO, PaddleOCR

**Development:** React, React Native, FastAPI, MongoDB, Git

## Leadership Experience

---

### KAIST MR (Microbot Research)

Daejeon, South Korea

Member

Since Mar 2025

### AI Robotics KR

South Korea

Community Manager

Oct 2024 – Sep 2025

Community Lead

Sep 2025 – Present

### Republic of Korea Army

Paju, South Korea

Sergeant; honorably discharged

Mar 2019 – Nov 2020

### Korean Red Cross Lifeguard

Daejeon, South Korea

Since Aug 2018

### KAIST Swimming Team KAORI

Daejeon, South Korea

Senior Member

Since Mar 2018

President of KAORI

2022

Chair, KAIST Swimming Competition

2023